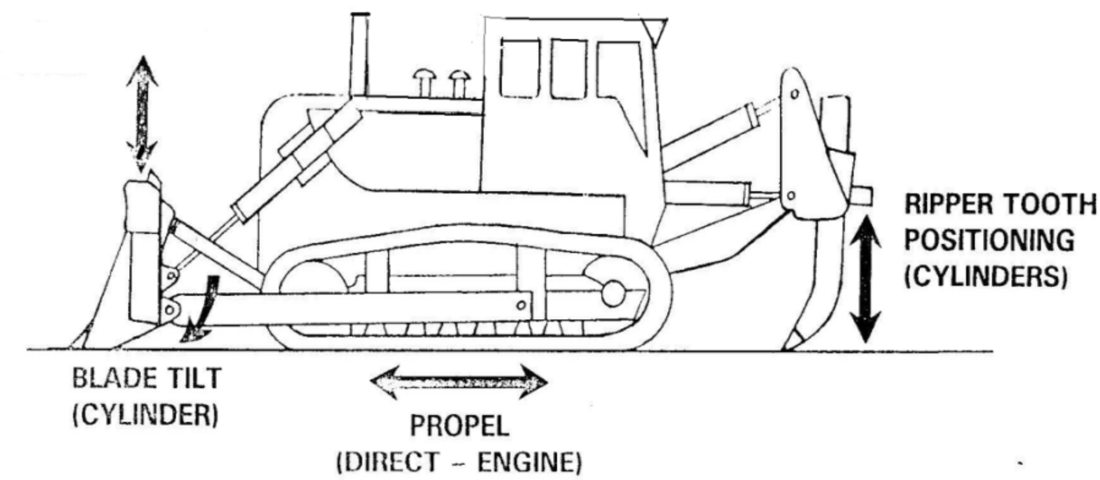
BULLDOZER

A bulldozer is a crawler mounted equipped with a metal plate in the front known as blade. It is used to push large quantities of soil, sand, rubble, or other such material during mining or construction work. When dozers are equipped at the rear with a claw shaped device it is known as a ripper. Rippers are being used to loosen rock material that are moderately hard.



Blades that are used in dozers may be classified into four groups

1. Straight Blade
2. Angling Blade
3. Push (Cushion) Blade
4. Universal Blade

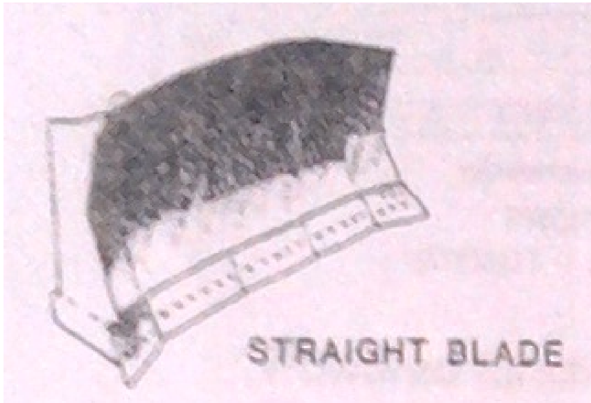
1. Straight Blade

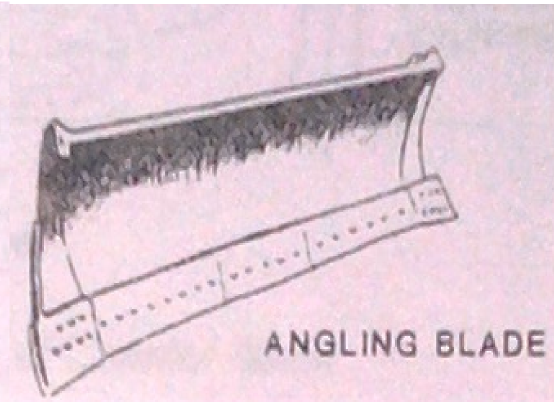
a) **Type -** Blade is fixed at right angle to the direction of travel. Blade can be raised / lowered below grade and the top edge can be pitched forwards or backwards by 5 to 10 deg.

b) **Selection criteria -** Performs best is in medium to harder compacted materials. It provides good penetration (pitched forward) and can handle heavy materials (pitched forwards).

c) **Application -** Backfilling, stumping, stripping, shaping and ditching.

d) **Limitation -** With the straight configuration and lack of side wings, to hinder side spillage of material, the blade has limited material carrying capabilities.



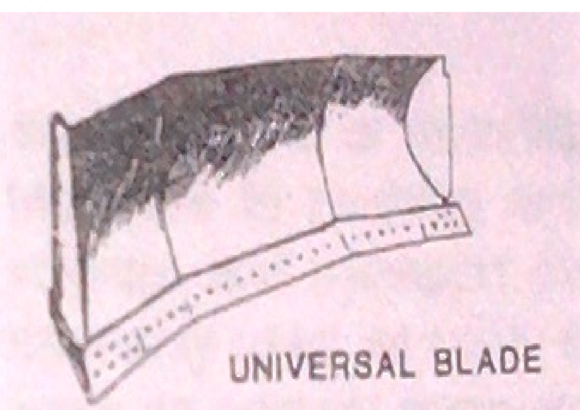


2. Angling Blade

a) **Type -** Blade is straight (long and narrow) and set at an angle to the direction of travel, often combined with a tilt feature. It is designed to side casting.
b) **Selection criteria –** It allows for greater operator ease of tilt adjustment and consequent productivity increases.
c) **Application -** Stumping and stripping, shaping, ditching, trench cutting, and general dozing of medium to softer materials.
d) **Limitation -** The blade allows material to easily spill off the sides thus reducing its value for carrying materials to longer distances.

3. Push (Cushion) Blade

a) **Type -** Blade is small to avoid scraper tire damage and is fixed at right angle to the direction of travel. Blade is cushioned at the middle. Blade can be raised / lowered below grade and the top edge can be pitched forwards or backwards.
b) **Selection criteria –** It is used to push scrapers (or trucks), has ability to clean up the cut area and increase the total fleet production.
c) **Application -** General dozing jobs.
d) **Limitation -** Not designed for production earth moving. It cannot be tilted, pitched, or angled. Allows material to easily spill off the sides thus reducing its value for carrying materials longer distances.



4. <u>Universal Blade</u>

a) **<u>Type –</u>** Blade has forward carving side and is fixed at right angle to the direction of travel. Blade is longer than the straight blade can be pitched.

b) **<u>Selection criteria –</u>** Designed to handle large loads over long distances for light and easily dozed material. Lowers the overall ground penetrating forces.

c) **<u>Application –</u>** Blade is suitable for medium to soft soils (i.e. lighter materials). Typical usages are working stockpiles and drifting loose or non cohesive materials.

d) **<u>Limitation -</u>** The cutting ratio* is lower for the Universal than the Straight blade mounted on a similar dozer. The Universal blade's load ratio* is lower than that of a similar Straight blade.

Apart from the above blade types, special blades may be designed for specific applications.

* cutting ratio (ratio of HP of dozer to cut length), load ratio (ratio of HP of dozer to volume of material cut)